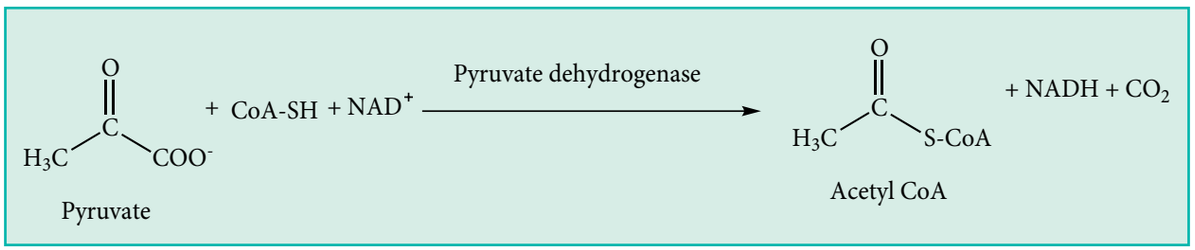

Step 1: In the first step of the cycle, the acetyl CoA (two-carbon) reacts with an oxaloacetate molecule (four-carbon) to form citrate (six-carbon) in a condensation reaction. This reaction is catalysed by the enzyme citrate synthase. The CoA is bound to a thiol group (-SH) and diffuses away to eventually combine with another acetyl group. This step is irreversible because it is highly exergonic. The rate of this reaction is controlled by negative feedback and the amount of ATP available. If ATP levels increase, the rate of this reaction decreases. If ATP is in short supply, the rate increases.

Step 2: In second step, citrate is converted into its isomer, isocitrate, by the enzyme aconitase.

Step 3: In the third step, the isocitrate is oxidized, to produce a five-carbon molecule, α-ketoglutarate releasing a molecule of CO_2. In this process two electrons are also released which reduce a NAD^+ molecule to NADH. This step is catalysed by isocitrate dehydrogenase.

Step 4: In the step four, α-ketoglutarate is oxidised to succinyl CoA a four carbon molecule in the prescence of Coenzyme A. In this step also a NAD^+ molecule is reduced to NADH. This reaction is catalysed by α-ketoglutarate dehydrogenase enzyme.

Step 5: In step five, the succinyl CoA is converted into succinate with the release of large amount of energy. This energy is utilised for the phosphorylation of a guanosine diphosphate (GDP) to guanosine triphosphate (GTP) by the addition of inorganic phosphate. GTP is energetically equivalent to ATP; it is primarily used in protein synthesis, however, its use is more restricted. This reaction is catalysed by the enyme, succinyl-CoA synthetase (succinate thiokinase).

Step 6: Step six is a dehydration process that converts succinate into fumarate. Two hydrogen atoms are transferred to FAD, producing $FADH_2$. Unlike NADH, this carrier remains attached to the enzyme succinate dehydrogenase which catalyses this reaction, and transfers the electrons to the electron transport chain directly.

Step 7: Water is added to fumarate in step seven, and malate is produced. This reaction is catalysed by fumarase.

Step 8: The last step of the citric acid cycle regenerates oxaloacetate by oxidizing malate. Another molecule of NADH is produced in this reaction, which is catalysed by malate dehydrogenase.

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